UNIVERSITY OF INFORMATION TECHNOLOGY
Faculty of Information Systems

Practical Exam
Subject: Object-Oriented Programming
Duration: 75 minute

Question 1: Square Class

Define a class Square to represent a square S with:

- **Vertices**: A(x1, y1), B(x2, y2), C(x3, y3), D(x4, y4)

Methods:

- area() → Calculate the area of the square
- perimeter() → Calculate the perimeter of the square
- checkEqualSides() → Check if the shape formed by the 4 points has four equal sides

Write the constructor and call all implemented methods in the main function.

Question 2: School Staff Management System

Context:
A school needs to manage its staff members, which include:

- Principal
- Teacher
- Administrator
- Janitor

Common Attributes:

- Staff ID
- Full Name
- Date of Birth
- Address
- Base Salary
- Date of Joining

Salary Calculation Rules:

- Principal: Salary = Base Salary × Coefficient
- Teacher: Salary = Base Salary + Teaching Hours × 100,000
- Administrator: Salary = Base Salary + Bonus
- Janitor: Salary = Base Salary + Night Shift Days × 50,000

Tasks:

1. Input a list of N staff members (must enter at least 4 records, and the types must be different — or can initialize the objects directly instead of inputting them from the keyboard).
2. Calculate and display the total salary of all staff members for the month.
3. If there is no Base Salary, compare the average salary difference between Teachers and Administrators (You may write this in the main method).